23 September 2019 **STA2020F TUT 9 MEMO** **NON-PARAMETRICS**

QUESTION 1

a) The Kruskal-Wallis test is appropriate here as we are dealing with four independent samples of quantitative data, and we want to compare them with respect to their location.

H_0: The locations (median) of all four groups are the same

H_1: At least two group's locations are different

Calculating the Test Statistic:

k = 4; n = 12

Ranking (combined sample):

Fear:	*8*	*12*	*2*		$T_1 = 22$
Happiness:	*7*	*10*	*1*	*5*	$T_2 = 23$
Depression:	*6*	*11*	*9*		$T_3 = 26$
Calmness:	*4*	*3*			$T_4 = 7$

Test Statistic:

$$H = \frac{12}{n(n+1)}\sum_{j=1}^{k}\frac{T_j^2}{n_j} - 3(n+1) = \frac{12}{12(13)}\left(\frac{22^2}{3} + \frac{23^2}{4} + \frac{26^2}{3} + \frac{7^2}{2}\right) - 3(13) = 2.801$$

p-value: $H \sim \chi_3{}^2$

p-value of the test statistic > 0.10

Conclusion: Since the p-value of the test statistic > 0.10 > 0.01 we DO NOT REJECT the null hypothesis, and conclude that the locations of all four groups are the same i.e. there is insufficient evidence to suggest that there is a difference in the skin potential of people experiencing different emotions.

b) No, it does not because not all the sample sizes are greater than or equal to 3. This does cast doubt on the validity of the test as it violates the assumption that the test statistic follows a chi-squared distribution.

QUESTION 2

The Spearman's rank correlation coefficient test is appropriate to conduct here as we are testing the degree of the relationship between two samples of ordinal data.

H_0: There is no association between the assessment scores across the two years

H_1: The assessment scores across the two years are associated

Test Statistic:

$n = 10$

This year:	5	6	4	5	5	4	7	6	4	5
Rank	*5.5*	*8.5*	*2*	*5.5*	*5.5*	*2*	*10*	*8.5*	*2*	*5.5*
Last year:	5	5	3	3	4	3	4	2	6	1
Rank	*8.5*	*8.5*	*4*	*4*	*6.5*	*4*	*6.5*	*2*	*10*	*1*
di	*-3*	*0*	*-2*	*1.5*	*1*	*-2*	*3.5*	*6.5*	*-8*	*4.5*

$$r_s = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)} = 1 - \frac{6(159)}{10(99)} = 1 - 0.964 = 0.036$$

$$z = r_s\sqrt{n-1} = 0.036\sqrt{9} = 0.108$$

Critical Region:

Reject H_0 if z > 1.96 or if z < -1.96

Conclusion:

Since z = -1.96 < 0.108 < 1.96 we do not Reject the null hypothesis, and can conclude that there is no assocation between the assessment scores across the two years, at the 5% significance level.

QUESTION 3

The Friedman test is appropriate here since we are comparing three related/dependent samples of ordinal data with respect to their locations.

H0: There are no differences in the ratings of the three recipes H1: There are differences in the ratings of the three recipes

Original	*Rank*	**New Recipe 1**	*Rank*	**New Recipe 2**	*Rank*
5	*2*	5	*2*	5	*2*
3	*1*	4	*2*	5	*3*
4	*1*	5	*2.5*	5	*2.5*
2	*1*	4	*2.5*	4	*2.5*
3	*1.5*	3	*1.5*	5	*3*
2	*1.5*	2	*1.5*	3	*3*
3	*2.5*	3	*2.5*	2	*1*
4	*2*	3	*1*	5	*3*
1	*2*	1	*2*	1	*2*
1	*1*	3	*3*	2	*2*
2	*1*	4	*3*	3	*2*
3	*1.5*	3	*1.5*	4	*3*
5	*3*	3	*1*	4	*2*
3	*2.5*	2	*1*	3	*2.5*
4	*2*	3	*1*	5	*3*
T1	25.5	**T2**	28	**T3**	36.5

Test statistic:

B = 15, k = 3

$$F_r = \left[\frac{12}{bk(k+1)}\sum_{j=1}^{k} T_j^2\right] - 3b(k+1)$$

$$F_r = 4.43$$

Critical region:

α=0.01

df =k−1=3−1=2

From tables, $\chi_2{}^2{}_{;0.01}$ = 9.210

Reject the null hypothesis if $F_r \geq 9.210$ or the approximate p-value < α=0.01

<u>Conclusion:</u>

We fail reject the null hypothesis since 4.43 < 9.21 and conclude that there are no significant differences in the ratings of the three recipes at the 1% significance level

OR

We do not reject the null hypothesis since the approximate p-value = 0.1 > 0.01 and conclude that there are no significant differences in the ratings of the three recipes at the 1% significance level